Project Title: malaria-detection using deep learning

Project Design Phase-I - Solution Fit

Team ID:

Define CS, fit into CC

1. CUSTOMER SEGMENT(S) CS

Who is your customer?: Medical laboratory technicians, pathologists, and healthcare workers in malaria-endemic regions.
Define CS: Includes diagnostic centers in rural areas with limited access to expert microscopists.
Focus: Targeting regions where manual inspection of 5,000+ cells per slide is a daily bottleneck.

6. CUSTOMER CONSTRAINTS CC

Spending Power: Low budget for high-end specialized diagnostic hardware.
Infrastructure: Need for a tool that can run on standard computing hardware (like a local PC or smartphone) without constant high-speed internet.

5. AVAILABLE SOLUTIONS AS

Manual Microscopy: The traditional gold standard, but labor-intensive.
RDTs (Rapid Diagnostic Tests): Faster but often less cost-effective or sensitive in certain resource-constrained settings.
PCR (Polymerase Chain Reaction): High performance but limited by cost and infrastructure requirements.

Explore AS, differentiate

Focus on J&P, tap into BE, understand RC

2. JOBS-TO-BE-DONE / PROBLEMS J&P

Jobs-to-be-Done: Automating the binary classification of blood cells into "Parasitized" or "Uninfected" categories.
Problems: Manual microscopy is time-consuming (takes intensive mental concentration), prone to human error (fatigue), and suffers from inter-observer variability.

9. PROBLEM ROOT CAUSE RC

Root Cause: A critical shortage of trained pathologists relative to the high prevalence of the disease globally (over 200 million cases annually).

7. BEHAVIOUR BE

Direct: Technicians manually scan blood smears under 100x magnification, counting cells one by one.
Indirect: Seeking second opinions from senior pathologists for ambiguous samples.

Focus on J&P, tap into BE, understand RC

Identify strong TR & EM

3. TRIGGERS TR

Triggers: Sudden outbreaks or high patient volumes in tropical regions.
Need: A diagnostic delay can lead to severe complications or death, creating an urgent need for fast, effective screening tools.

4. EMOTIONS: BEFORE / AFTER EM

Before: Feeling overwhelmed by the volume of slides and anxious about missing a life-threatening infection due to fatigue.
Insert text here
After: Feeling confident in the consistency of digital results and relieved by the significantly reduced workload.

10. YOUR SOLUTION SL

Solution: A Convolutional Neural Network (CNN) model that processes cell images to provide high-accuracy (e.g., ~95%+) automated detection of malaria parasites.

8. CHANNELS of BEHAVIOUR CH

Online: Utilizing medical image repositories (like NIH) and open-source GitHub tools for diagnostic assistance.
Offline: Working in local clinics, pathology labs, and public health centers.

Identify strong TR & EM